PSET 5: Functions of several variables and optimization with several variables

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Find first partial derivatives

Find all of the first partial derivatives of each function.[1]

a. $f(x, y) = \frac{1}{3}x^5 + 3x^3y^2 + 3xy^4$

b. $g(x, y) = 4xe^{4y+1}$

c. $k(x, y) = \dfrac{x - y}{x + y}$

Problem 2: Find the gradient, Hessian and Jacobian

Find the gradient, Hessian H, and Jacobian, J, for the following functions.[2] Evaluate the gradient at the given points.

a. $g(x, y) = x^4 + y^4 - 4xy + 2$

   i. Gradient: (evaluate at: $(x, y) = (1, 1)$)

   ii. Jacobian:

   iii. Hessian:

b. $k(x, y) = (1 + xy)(x + y)$

   i. Gradient: (evaluate at: $(x, y) = (1, -1)$)

   ii. Jacobian:

   iii. Hessian:

[1] Inspired by Grimmer HW6.3

[2] Inspired by Grimmer HW7.3

Problem 3: Find the critical points

Find the local minimum values, local maximum values, and saddle point(s) of the function. Remember the process we discussed in class: calculate the gradient, set it equal to zero to solve the system of equations, calculate the Hessian, and assess the Hessian at critical values. Be sure to show your work on each of these steps.[3]

a. $h(x, y) = x^4 + y^4$

b. $g(x, y) = x^4 + y^4 - 4xy + 2$ (You can reuse your calculations from above)

c. $k(x, y) = (1 + xy)(x + y)$ (You can reuse your calculations from above)

Problem 4: Definite and Indefinite integrals

The remaining problems cover integration. Solve the following integrals using the antiderivative method.[4]

For all these problems, the basic approach to compute the definite integral of $f(x)$ from a to b is by using the formula $F(b) - F(a)$, where $F(x)$ is the **antiderivative** of f.

a. $\int_{-1}^{0} (3x^2 - 3)\, dx$

b. $\int_{2}^{4} e^{3y}\, dy$

c. $\int_{3}^{3} \sqrt{x^5 - 2}\, dx$

d. $\int \left(x^2 - x^{-\frac{1}{2}}\right) dx$

Problem 5: Substitution and integration by parts

a. $\int 2x(x^2 + 1)^4\, dx$

b. $\int \dfrac{3x^2}{x^3 + 5}\, dx$

c. $\int xe^{2x}\, dx$

d. Without solving, state whether you would use substitution or integration by parts for each, and why:

   i. $\int 5x^4 \left(x^5 - 2\right)^3 dx$

   ii. $\int x^2 \log(x)\, dx$

[3]Inspired by Grimmer HW7.4

[4]Inspired by Gill 5.10, Grimmer HW4.1, Gill 5.13, and 5.14

Problem 6: Improper and multiple integrals

For (a)–(c), determine whether the integral is convergent or divergent, and evaluate those that are convergent.[5] For (d)–(e), evaluate the iterated integral. In each case, state whether the order of integration is forced, and why.

a. $\int_1^\infty \left(\frac{1}{3x}\right)^2 dx$

b. $\int_0^\infty e^{-x}\, dx$

c. $\int_{-\infty}^0 x^3\, dx$

d. $\int_0^1 \int_2^3 x^2 y^3\, dx\, dy$

e. $\int_0^1 \int_0^{\sqrt{1-x^2}} 2x^3 y\, dy\, dx$

Problem 7: Interpretation

No new computation is required for this problem. Answer each part in two to four sentences of prose.

a. **Reading a partial derivative.** Suppose we regress presidential approval in month i on the employment rate and the price of gas, and obtain

$$\text{Approval}_i = 0.8 + 0.5\,\text{Employ}_i - 0.25\,\text{Gas}_i.$$

Write down $\dfrac{\partial\,\text{Approval}_i}{\partial\,\text{Employ}_i}$ and interpret it substantively. Your interpretation should make clear what is happening to Gas_i while employment changes, and why that matters for the claim you are making.

b. **Gradient versus Hessian.** Explain why the condition $\nabla f(\mathbf{a}) = \mathbf{0}$ is *necessary* but not *sufficient* for $\mathbf{a}$ to be a local extremum. Use your answers to Problem 3 to illustrate: both $(0,0)$ and $(1,1)$ have zero gradient, but they are different kinds of point. What does the Hessian tell you at each that the gradient could not? Finally, explain what it means when the determinant test returns $AC - B^2 = 0$, and why we cannot classify the point in that case.

c. **Integrals and probability.** A continuous probability density f must satisfy $\int_{-\infty}^\infty f(x)\, dx = 1$. Explain what this requirement means, and what $\int_a^b f(x)\, dx$ represents for some interval $[a, b]$.

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for

[5] Inspired by Grimmer HW4.3

help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.